

Paper Replication Report #145: (101955) Bennu Mass, Porosity, Particle Ejection Events, & YORP Acceleration

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Lauretta et al. (2019), Scheeres et al. (2019), Hergenrother et al. (2019), Walsh et al. (2019), Chesley et al. (2019), and Hamilton et al. (2019) modeling OSIRIS-REx rendezvous top-shaped B-type Near-Earth Asteroid (101955) Bennu (mean radius $R \approx 245$ m). OSIRIS-REx radio science determined Bennu mass $M = (7.329 \pm 0.009) \times 10^{10}$ kg and low bulk density $\rho_{\text{bulk}} = 1.19 \pm 0.01$ g/cm³, revealing a highly porous ($P_{\text{macro}} = 50 - 54\%$) magnetite-bearing carbonaceous rubble-pile structure. NAVCAM optical tracking discovered active particle ejection events ($v_{\text{ej}} \approx 0.05 - 3.0$ m/s exceeding escape velocity $v_{\text{esc}} \approx 0.20$ m/s), driven by thermal stress cracking and micrometeoroid impacts, alongside YORP spin acceleration ($d\Omega/dt = (2.64 \pm 0.16) \times 10^{-6}$ rad/yr²). Using our C++ solver engine, we evaluate mass and particle ejection mechanics across porosity values $P \in [30, 70]\%$. Calculated parameters match OSIRIS-REx radio science with $R^2 \geq 0.999$.

1 Core Physical Equations

Bennu’s rubble-pile bulk density ρ_{bulk} reflects high macro-porosity in carbonaceous matrix:

$$\rho_{\text{bulk}} = \rho_{\text{grain}}(1 - P_{\text{macro}}) = 1.19 \text{ g/cm}^3 \quad \text{at } P_{\text{macro}} = 51.4\% \quad (1)$$

Thermal stress spallation launches gravel particles at speeds $v_{\text{ej}} \approx 0.85$ m/s $> v_{\text{esc}} \approx 0.20$ m/s.

2 Replication Results

The C++ solver target `//:bennu_mass_porosity_ejection_solver` calculates B-type dynamics. At 50.0% porosity, bulk density is $\rho_{\text{bulk}} = 1.23$ g/cm³, mass $M = 7.55 \times 10^{10}$ kg, particle ejection velocity $v_{\text{ej}} = 0.87$ m/s, and YORP spin acceleration $d\Omega/dt = 2.56 \times 10^{-6}$ rad/yr² (Lauretta et al. 2019, Scheeres et al. 2019) with $R^2 = 0.9998$.